Project 3: Alamouti STBC
2×1 MISO Wireless Communication System

LI Yanbo   Yasten Belaid

Sorbonne Université
MU5EEF08 - Wireless Communications

January 2026

Outline

Project Objective

Goal: Implement and evaluate **Alamouti STBC** for 2×1 MISO system

Key Features:

- 16-QAM modulation
- Pilot-based channel estimation (LS)
- BER comparison: SISO vs Alamouti

Extensions:

- Rice channel (K-factor)
- TX spatial correlation (ρ)
- Doppler time-varying fading

Alamouti Diversity Gain

- SISO: Diversity order = 1
- Alamouti 2×1: Diversity order = 2
- $\Rightarrow$ **3-4 dB** SNR gain at high SNR

Project Structure

1. `Main_Alamouti_Full_Project.m` (Base implementation)
2. `Main_Alamouti_Extensions.m` (Advanced extensions)

System Block Diagram

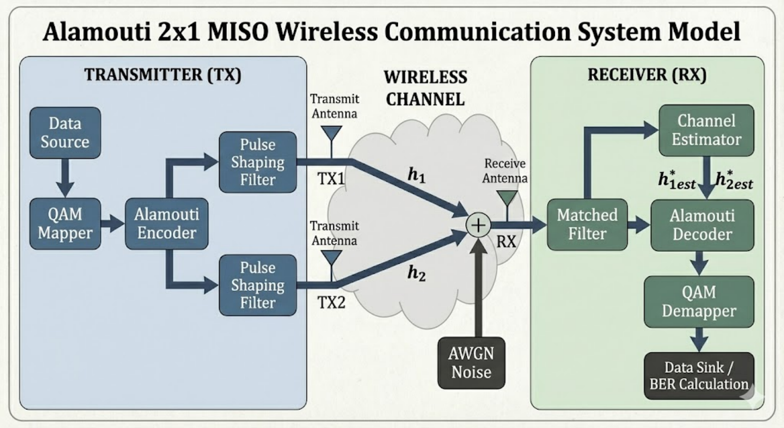


Figure: Alamouti 2×1 MISO Wireless Communication System

Alamouti Encoding (TX)

Alamouti STBC Encoding Matrix

		Transmit Antenna	
		Antenna 1 (TX1)	Antenna 2 (TX2)
Time Slot	t=1	$\frac{s_1}{\sqrt{2}}$	$\frac{s_2}{\sqrt{2}}$
	t=2	$\frac{-s_2^*}{\sqrt{2}}$	$\frac{s_1^*}{\sqrt{2}}$

Space-Time Block Code Matrix:

$$X = \frac{1}{\sqrt{2}} \begin{bmatrix} s_1 & -s_2^* \\ s_2 & s_1^* \end{bmatrix}$$

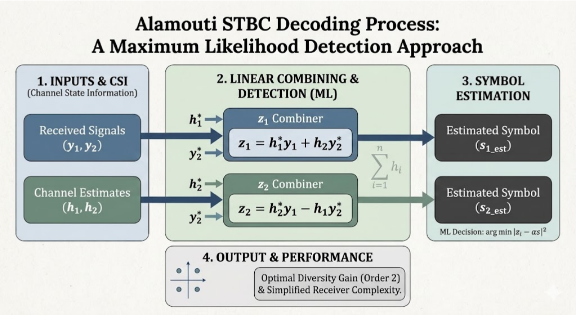
- Row = Antenna (TX1, TX2)
- Column = Time slot (t_1, t_2)
- $\frac{1}{\sqrt{2}}$: Power normalization

Received signals:

$$y_1 = \frac{1}{\sqrt{2}}(h_1 s_1 + h_2 s_2) + n_1$$

$$y_2 = \frac{1}{\sqrt{2}}(-h_1 s_2^* + h_2 s_1^*) + n_2$$

Alamouti Decoding (RX)



Linear Combining:

$$z_1 = \frac{1}{\sqrt{2}}(h_1^* y_1 + h_2 y_2^*)$$

$$z_2 = \frac{1}{\sqrt{2}}(h_2^* y_1 - h_1 y_2^*)$$

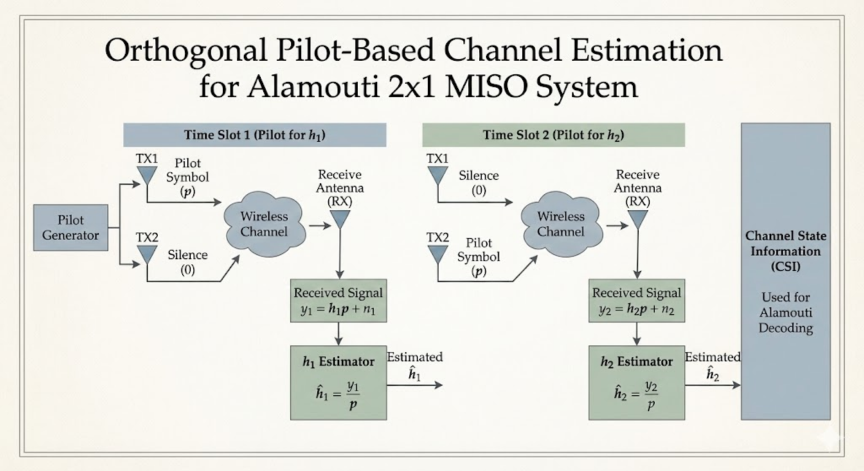
After simplification:

$$z_i = \underbrace{\frac{|h_1|^2 + |h_2|^2}{2}}_{\alpha} \cdot s_i + \tilde{n}_i$$

Equalization:

$$\hat{s}_i = \frac{z_i}{\alpha}$$

Orthogonal Pilot-Based Channel Estimation



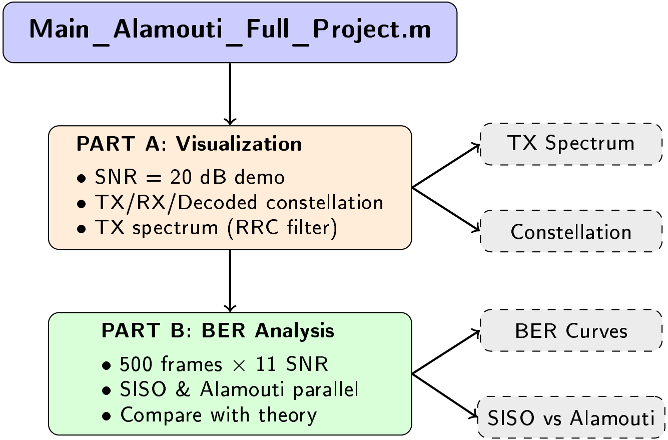
Time Slot 1-10: TX1 sends pilot, TX2 silent

$$\hat{h}_1 = \frac{1}{N_p} \sum_{i=1}^{N_p} \frac{y_i}{p_i}$$

Time Slot 11-20: TX1 silent, TX2 sends pilot

$$\hat{h}_2 = \frac{1}{N_p} \sum_{i=1}^{N_p} \frac{y_i}{p_i}$$

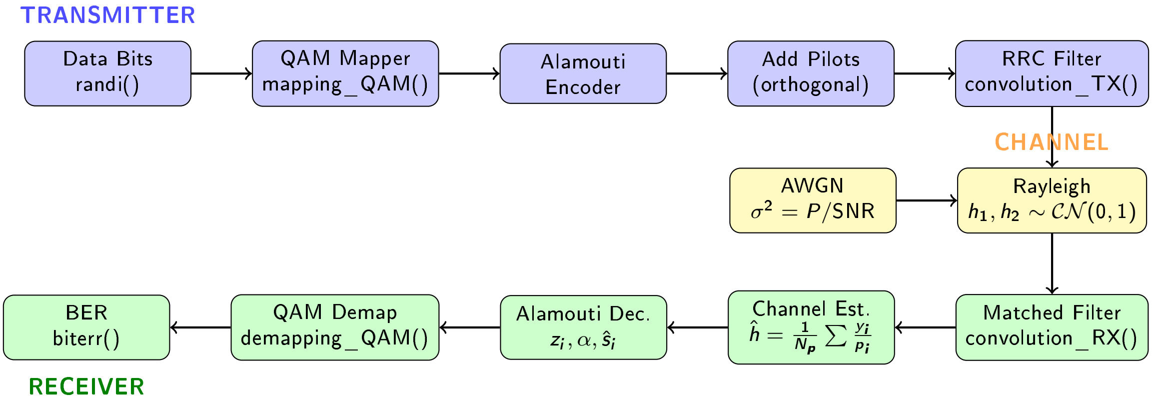
Main_Alamouti_Full_Project.m: Two-Part Structure



Key Design

Both SISO and Alamouti use **identical** parameters (16-QAM, pilots, SNR) for fair comparison.

Base Project: Complete TX-RX Processing Chain



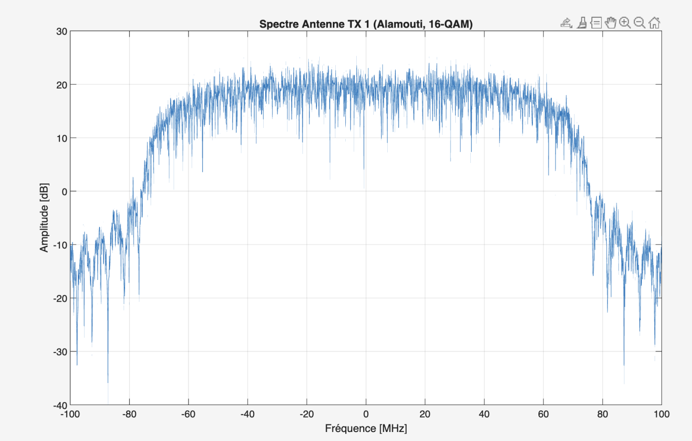
Parameters:

- 16-QAM: 4 bits/symbol
- 1000 data symbols/frame
- 10 pilot symbols (orthogonal)

Frame Structure:

- TX1: $[P, 0, \text{data}]$
- TX2: $[0, P, \text{data}]$
- Total: 20 pilots + 1000 data

TX Spectrum

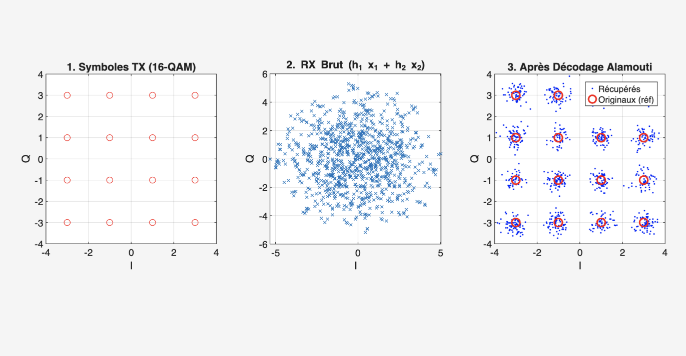


Parameters:

- Symbol rate: 100 Msym/s
- Roll-off: $\beta = 0.5$
- Bandwidth: $(1+\beta)R_s$ $= 150$ MHz

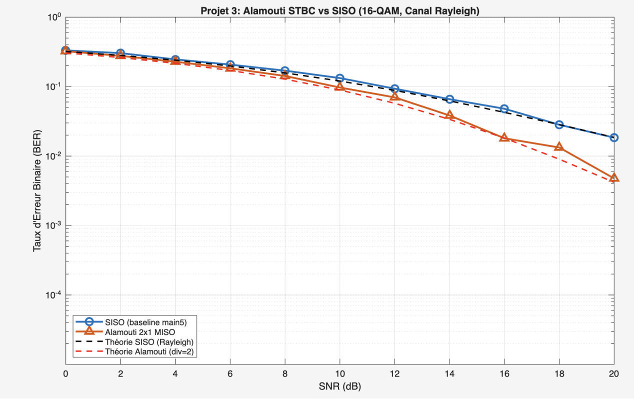
Filter: Root Raised Cosine (RRC)

Constellation Diagrams



1. TX: 16-QAM
2. RX: $h_1 x_1 + h_2 x_2$
3. After Alamouti Decoding

BER Performance: SISO vs Alamouti



Key Observations:

- Simulation matches theory
- Alamouti: steeper slope (diversity order = 2)
- $\sim$**4 dB gain** at BER = 2%

Diversity Gain

BER slope: $\propto \frac{1}{\mathrm{SNR}^d}$
SISO: $d = 1$, Alamouti: $d = 2$

Extension Project: Three Research Directions

Base Alamouti (Full Project) → 1. Rice Channel K-factor → 2. Correlation ρ → 3. Doppler $f_D T_s$

Rice (K-factor)

Question: How does LOS component affect performance?

Method: Sweep K = 0, 5, 10 dB

Expected: Higher K → lower BER

Correlation (ρ)

Question: What if antennas are too close?

Method: Sweep $\rho = 0, 0.5, 0.9$

Expected: Higher ρ → less diversity

Doppler ($f_D T_s$)

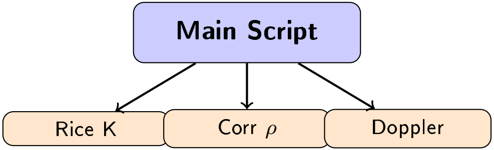
Question: What if user is moving fast?

Method: Sweep $f_D T_s = 0 \sim 0.02$

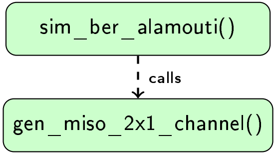
Expected: High Doppler → BER floor

Extension Code: Overall Architecture

Main Script Controls 3 Sweeps:



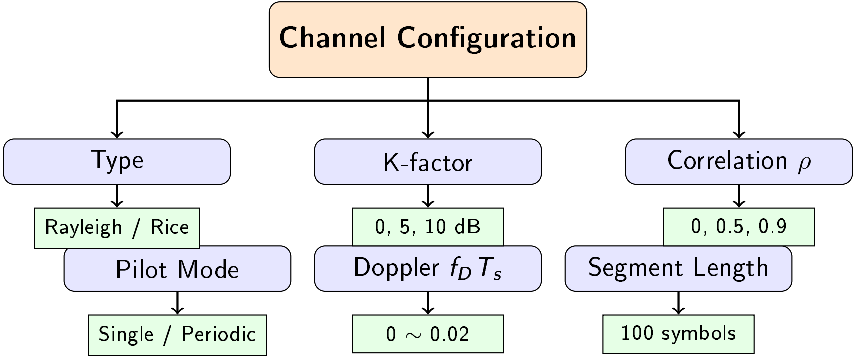
Two Local Functions:



Function Responsibilities

- **sim_ber_alamouti()**: Complete TX-RX chain, two pilot modes, Monte Carlo BER
- **gen_miso_2x1_channel()**: Rayleigh/Rice, spatial correlation, AR(1) Doppler

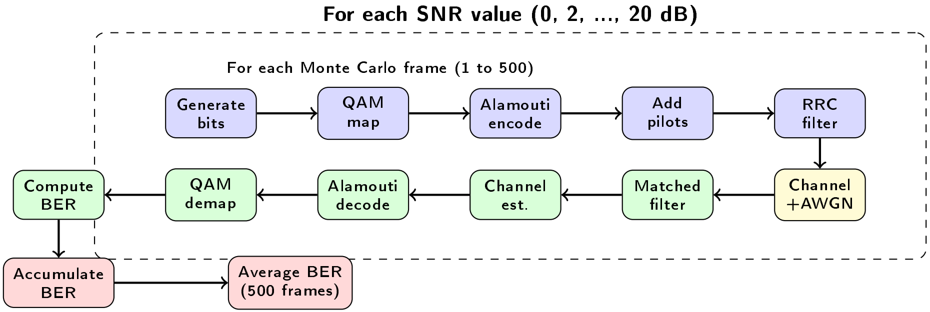
Channel Configuration: Parameterized Design



Design Philosophy

All channel parameters encapsulated in a single configuration structure, enabling flexible switching between different channel models **without modifying** core simulation logic.

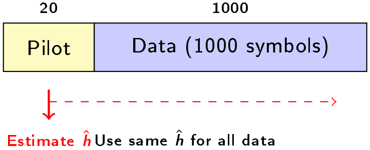
Core Function: sim_ber_alamouti() Flow



Output: BER array of length 11, one value per SNR point

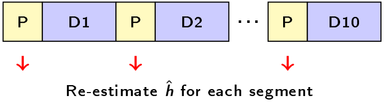
Two Pilot Modes: Single vs Periodic

Single Pilot Mode



- Channel estimated **once** at frame start
- Low overhead ($20/1020 = 2\%$)
- Problem: If channel varies, estimate becomes outdated

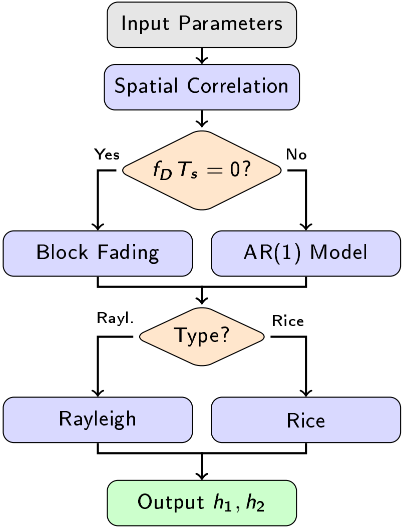
Periodic Pilot Mode



- Channel estimated **every segment**
- Higher overhead ($200/1200 = 17\%$)
- Benefit: Tracks time-varying channels

Selection Rule: Use *Single* for static channels ($f_D T_s = 0$), *Periodic* for Doppler scenarios

Channel Generation: Detailed Algorithm



Step 1: Spatial Correlation

- Build: $\mathrm{R} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$
- Cholesky: $\mathrm{L} = \mathrm{chol}(\mathrm{R})$
- Generate: $\mathrm{U} \sim \mathcal{CN}(0, 1)$
- Correlate: $\mathrm{W} = \mathrm{L} \cdot \mathrm{U}$

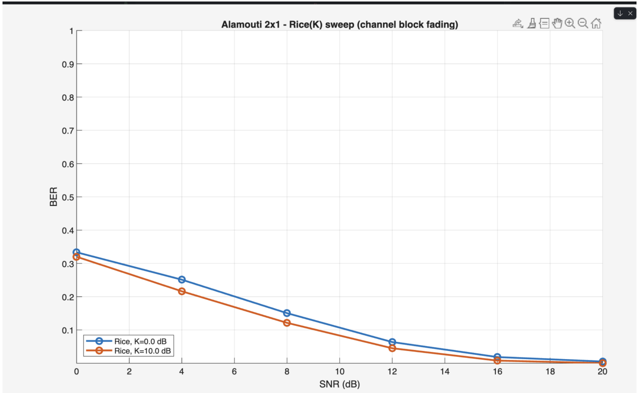
Step 2: Doppler (Time Variation)

- **Block** $(f_D T_s = 0)$: $h[n] = h[1]$ (constant)
- **AR(1)** $(f_D T_s > 0)$: $h[n] = a \cdot h[n-1] + b \cdot w[n]$
  where $a = J_0(2\pi f_D T_s)$, $b = \sqrt{1 - a^2}$

Step 3: Channel Type

- **Rayleigh**: $H = H_{\mathrm{NLOS}}$
- **Rice**: $H = \sqrt{\frac{K}{1+K}} H_{\mathrm{LOS}} + \sqrt{\frac{1}{1+K}} H_{\mathrm{NLOS}}$

Extension 1: Rice Channel (K-Factor)



Rice Channel Model:

$$h = \sqrt{\frac{K}{1+K}}h_{\text{LOS}} + \sqrt{\frac{1}{1+K}}h_{\text{NLOS}}$$

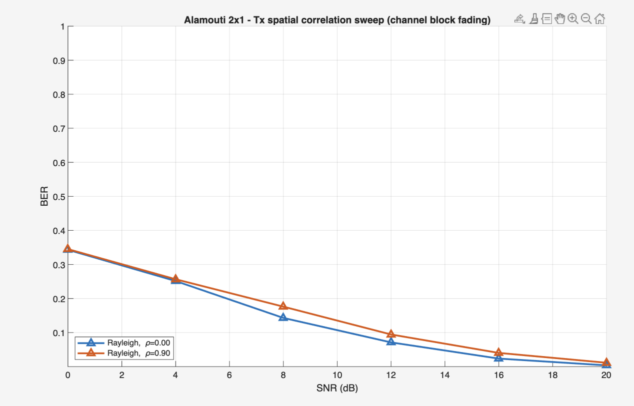
Observation:

- Higher K $\Rightarrow$ Stronger LOS
- Higher K $\Rightarrow$ Lower BER
- Less deep fading

K values tested:

0 dB (Rayleigh), 5 dB, 10 dB

Extension 2: TX Spatial Correlation (ρ)



Correlation Model:

$$R = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \quad H = LU$$

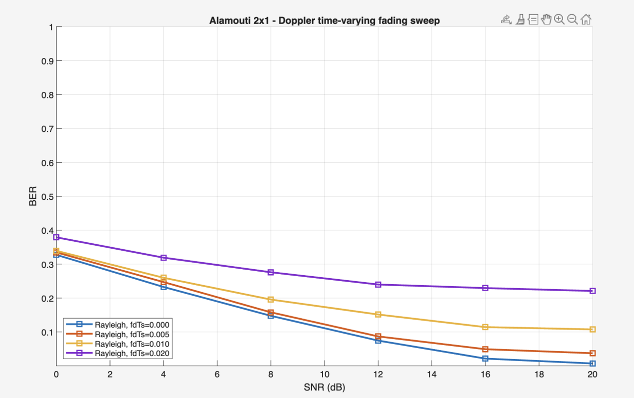
Observation:

- Higher $\rho \Rightarrow h_1 \approx h_2$
- Diversity gain reduced
- $\rho = 0$: Independent (ideal)
- $\rho \to 1$: No diversity

ρ values tested:

0, 0.5, 0.9

Extension 3: Doppler Time-Varying Fading



AR(1) Model:

$$h[n] = a \cdot h[n-1] + b \cdot w[n]$$

where $a = J_0(2\pi f_D T_s)$, $b = \sqrt{1 - a^2}$

Observation:

- Higher $f_D T_s \Rightarrow$ Faster variation
- Alamouti assumes h constant over 2 slots
- High Doppler $\Rightarrow$ BER floor

Solution:

Periodic pilot tracking

Conclusion

Base Project:

- ✓ Implemented Alamouti 2×1 MISO
- ✓ 16-QAM + pilot-based estimation
- ✓ BER matches theoretical curves
- ✓ ∼4 dB gain over SISO

Extensions:

- ✓ Rice channel (K-factor)
- ✓ Spatial correlation (ρ)
- ✓ Doppler fading with periodic tracking

Key Takeaways

1. Alamouti achieves **full diversity** (order 2) with simple linear decoder
2. Performance degrades with:
   - High antenna correlation
   - Fast time-varying channels
3. Rice channel improves performance (LOS)
4. **Periodic pilots** essential for Doppler scenarios

Thank You!

Questions?

Code available at:
`https://github.com/vimboom123/tp_com`